Key Challenges Faced by Data Engineers in the Project

Problem 1 – Inadequate Lower Environment Testing

Situation / Impact:

- Dev/QA shared infrastructure with limited and non■representative data volumes.
- Unable to validate transformations and performance fully before production.
- Production defects and slow jobs observed post deployment.

Mitigation:

- Generated synthetic and large test datasets.
- Implemented unit testing using pytest for PySpark jobs.
- Added performance and load testing before releases.

Problem 2 – Upstream Schema Drift & Data Quality Issues

Situation / Impact:

- Frequent schema changes without prior communication.
- Unexpected delays and malformed data from source systems.
- Pipelines failed due to type mismatches and missing fields.

Mitigation:

- Established data contracts and SLAs.
- Handled schema evolution using dynamic ingestion patterns.
- Added validation, alerts, and quarantine tables.

Problem 3 – Shared Cluster Resource Contention

Situation / Impact:

- Multiple teams sharing the same compute clusters.
- Job starvation and unpredictable runtimes.

Mitigation:

- Created dedicated queues and resource pools.

- Separated heavy workloads and scheduled off■peak runs.

Problem 4 – Cloud Resource Overuse & Cost Spikes

Situation / Impact:

- Oversized clusters and warehouses.
- Idle resources increased costs.

Mitigation:

- Right■sized compute, enabled auto■scaling and auto■suspend.
- Implemented tagging and cost monitoring dashboards.

Problem 5 – Data Skew

Situation / Impact:

- Uneven key distribution caused long-running tasks and executor failures.

Mitigation:

- Used repartitioning, key salting, broadcast joins, and optimized partitions.

Problem 6 – Limited Business SME Support

Situation / Impact:

- Insufficient domain knowledge for clarifications.
- Delayed troubleshooting and rework.

Mitigation:

- Documented business rules and conducted knowledge-sharing sessions.
- Scheduled regular sync meetings with stakeholders.

Problem 7 – Observability & Monitoring Gaps

Situation / Impact:

- Limited logging made debugging difficult.

Mitigation:

- Implemented centralized logging, dashboards, SLA alerts, and lineage tracking.

Problem 8 – Dependency Management Across Pipelines

Situation / Impact:

- Manual job sequencing caused failures and partial loads.

Mitigation:

- Adopted orchestration with DAGs and built idempotent jobs.

Problem 9 – Manual Deployments

Situation / Impact:

- Manual releases increased errors and slowed delivery.

Mitigation:

- Implemented CI/CD pipelines and automated testing.